1 Midterm Review

Midterm Review: This document covers key exam-style questions spanning Lectures 1–6. For each problem, we recall the relevant definitions and theorems from class, provide visual intuition, and outline a proof strategy—without writing the full proof.

Tips for the Exam:

- You do **not** have to prove a given hint unless otherwise specified.
- You do **not** have to do the problems in order.
- Shorthand notations are fine (e.g. $\forall$, $\exists$, $\Rightarrow$, $\Leftrightarrow$) as long as your meaning is clear.
- You may use **any equivalent definition** as long as it is correct.
- **1 page** printed/typed cheat sheet allowed.

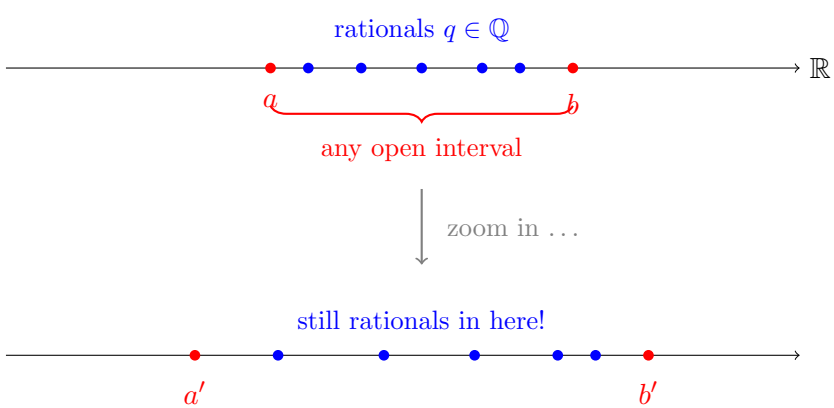
1.1 Question 1: "$\mathbb{Q}$ is dense in $\mathbb{R}$" and "$\mathbb{R}$ is the completion of $\mathbb{Q}$"

Question: Explain the meaning of "$\mathbb{Q}$ is dense in $\mathbb{R}$" and "$\mathbb{R}$ is the completion of $\mathbb{Q}$."

Part 1: "$\mathbb{Q}$ is dense in $\mathbb{R}$"

Definition (Density of $\mathbb{Q}$ in $\mathbb{R}$, Lecture 2). $\mathbb{Q}$ is **dense** in $\mathbb{R}$ means: for every $a, b \in \mathbb{R}$ with $a < b$, there exists $q \in \mathbb{Q}$ such that $a < q < b$.

In other words, *between any two distinct real numbers, no matter how close together, there is always a rational number sitting between them.* There is no "gap" in $\mathbb{R}$ that is free of rationals.



No matter how far you zoom in on the number line, you will always find rational numbers. You can never isolate an interval of pure irrationals.

How density is proved (Lecture 2). The proof relies on the **Archimedean property**: for any $x, y \in \mathbb{R}$ with $x > 0$, there exists $n \in \mathbb{N}$ with $nx > y$.
The strategy:

1. Since $b - a > 0$, apply the Archimedean property to find $n \in \mathbb{N}$ with $n(b-a) > 1$.
2. This means $nb - na > 1$, so there is an integer m with $na < m < nb$.
3. Then $q = \frac{m}{n} \in \mathbb{Q}$ satisfies $a < q < b$.

The key insight: the Archimedean property lets you "scale up" any tiny interval (a, b) until it is wide enough (width > 1) to guarantee an integer lands inside. Then you scale back down to get a rational.

Equivalent topological viewpoint. In Lecture 4, we defined the **closure** of a set E as the smallest closed set containing E. Equivalently, $x \in \overline{E}$ if and only if every neighborhood of x intersects E. Saying "$\mathbb{Q}$ is dense in $\mathbb{R}$" is the same as saying:

$$\overline{\mathbb{Q}} = \mathbb{R}$$

That is, every real number is either rational or a limit point of the rationals. For any $x \in \mathbb{R}$ and any $\varepsilon > 0$, the ball $B(x, \varepsilon) = (x - \varepsilon, x + \varepsilon)$ contains a rational number.

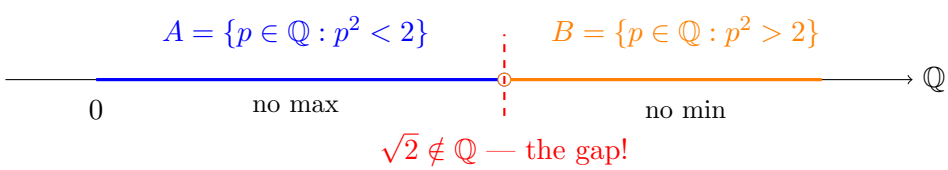
Part 2: "$\mathbb{R}$ is the completion of $\mathbb{Q}$"

This phrase captures a different idea: $\mathbb{Q}$ has "gaps" (missing points), and $\mathbb{R}$ is what you get when you fill them all in.

The gap in $\mathbb{Q}$ (Lecture 1). Consider the sets

$$A = \{p \in \mathbb{Q} : p > 0,\ p^2 < 2\}, \qquad B = \{p \in \mathbb{Q} : p > 0,\ p^2 > 2\}.$$

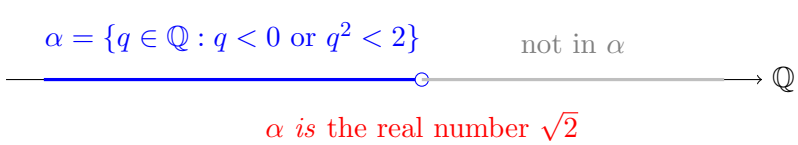
We proved that A has no largest element and B has no smallest element. The "number" $\sqrt{2}$ that should separate them simply does not exist in $\mathbb{Q}$. So A is bounded above in $\mathbb{Q}$, yet $\sup A$ does not exist in $\mathbb{Q}$.



This is precisely the failure of the **Least Upper Bound Property** in $\mathbb{Q}$: there exists a nonempty, bounded-above subset of $\mathbb{Q}$ whose supremum does not exist in $\mathbb{Q}$.

Definition (LUBP, Lecture 1). An ordered set S has the **Least Upper Bound Property** if every nonempty subset $E \subseteq S$ that is bounded above has $\sup E \in S$.

How $\mathbb{R}$ fills the gaps (Lecture 2). We construct $\mathbb{R}$ via **Dedekind cuts**: a cut is a nonempty, proper, downward-closed subset $\alpha \subset \mathbb{Q}$ with no maximum. Each cut represents a real number by encoding "everything to its left."



The set of all Dedekind cuts forms an ordered field $\mathbb{R}$ that:

1. **Contains $\mathbb{Q}$:** each rational p corresponds to the cut $p^* = \{q \in \mathbb{Q} : q < p\}$.
2. **Has the LUBP:** for any nonempty bounded-above collection of cuts, $\sup E = \bigcup_{\alpha \in E} \alpha$ is again a cut (Lecture 2).
3. **Is minimal:** $\mathbb{Q}$ is dense in $\mathbb{R}$, so $\mathbb{R}$ adds no "extra" points beyond what is needed to fill the gaps.

Summary: density vs. completion.

	"$\mathbb{Q}$ is dense in $\mathbb{R}$"	**"$\mathbb{R}$ is the completion of $\mathbb{Q}$"**
Says	Between any two reals there is a rational.	$\mathbb{R}$ is $\mathbb{Q}$ with all "gaps" filled in.
Formal	$\overline{\mathbb{Q}} = \mathbb{R}$ (closure in the metric topology).	$\mathbb{R}$ is an ordered field containing $\mathbb{Q}$ that satisfies the LUBP.
Key tool	Archimedean property.	Dedekind cuts (or Cauchy sequences).
Direction	Starts from $\mathbb{R}$, looks "down" at $\mathbb{Q}$ inside it.	Starts from $\mathbb{Q}$, builds "up" to $\mathbb{R}$.

Density says $\mathbb{Q}$ is *everywhere* in $\mathbb{R}$. Completion says $\mathbb{R}$ is the *smallest* ordered field containing $\mathbb{Q}$ with no gaps (i.e., satisfying the LUBP). Together: $\mathbb{R}$ is built from $\mathbb{Q}$ by filling in the missing suprema, and the original rationals remain spread throughout.

1.2 Question 2: Open vs. Closed Sets; Characterizing Open Sets in Metric Spaces

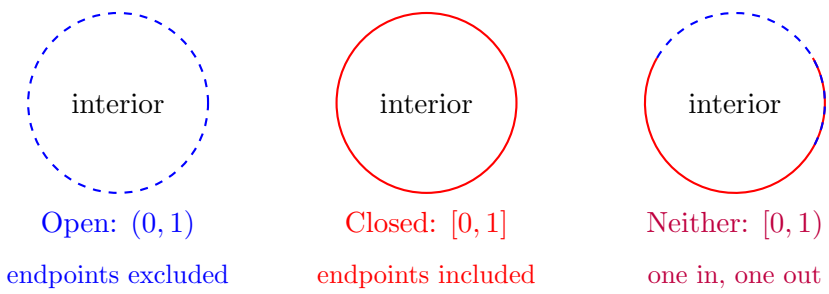
Question: What is the relation between open and closed sets? Characterize open sets in metric spaces.

Open and Closed Sets are Complements

Definition (Lecture 4). Let X be a topological space with topology $\mathcal{T}$.

- A set $U \subseteq X$ is **open** if $U \in \mathcal{T}$.
- A set $C \subseteq X$ is **closed** if its complement $X \setminus C$ is open.

The relationship is definitional: *closed means complement is open.* They are dual notions, not opposites. A set can be both open and closed (e.g., $\emptyset$ and X), or neither (e.g., $[0, 1)$ in $\mathbb{R}$).



Dual Properties (Lecture 4)

The axioms for open and closed sets are mirrors of each other:

Open sets	**Closed sets**
$\emptyset$ and X are open	$\emptyset$ and X are closed
Arbitrary unions of open sets are open	**Arbitrary** intersections of closed sets are closed
Finite intersections of open sets are open	**Finite** unions of closed sets are closed

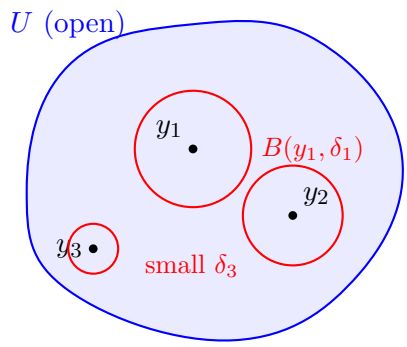
The switch from "union $\leftrightarrow$ intersection" and "arbitrary $\leftrightarrow$ finite" comes from De Morgan's laws: $X \setminus \bigcup U_\alpha = \bigcap (X \setminus U_\alpha)$.

Why only finite intersections for open sets? Consider $\bigcap_{n=1}^{\infty} (-\frac{1}{n}, \frac{1}{n}) = \{0\}$. Each $(-\frac{1}{n}, \frac{1}{n})$ is open, but their intersection is $\{0\}$, which is *not* open in $\mathbb{R}$ (no ball around 0 is contained in $\{0\}$). So infinite intersections of open sets can fail to be open.

Characterization of Open Sets in Metric Spaces

Theorem (Lecture 4). Let (X, d) be a metric space. A set $U \subseteq X$ is open if and only if for every $y \in U$, there exists $\delta > 0$ such that $B(y, \delta) \subseteq U$.

In words: U is open exactly when every point of U has "room around it" that stays inside U.



Every point has a ball fitting inside U.
Points near the boundary get smaller balls.

Proof strategy (Lecture 4).

- ($\Rightarrow$) Follows directly from the definition of the topology generated by a basis (open balls are the basis elements).
- ($\Leftarrow$) For each $y \in U$, pick $\delta_y > 0$ with $B(y, \delta_y) \subseteq U$. Then $U = \bigcup_{y \in U} B(y, \delta_y)$. Each ball is a basis element (hence open), and an arbitrary union of open sets is open.

This characterization is the bridge between the abstract topology axioms and concrete ε-δ arguments.

1.3 Question 3: Interior and Closure of a Set

Question: Define the interior and closure of a set.

Closure (Lecture 4)

Definition. The **closure** of a set $E \subseteq X$ is the smallest closed set containing E:

$$\overline{E} = \bigcap \{C \subseteq X : C \text{ is closed and } E \subseteq C\}.$$

Equivalent characterization (Lecture 4): $x \in \overline{E}$ if and only if every neighborhood of x intersects E. That is,

$$x \in \overline{E} \iff \text{for all open } U \ni x, \quad U \cap E \neq \emptyset.$$

Decomposition (Lecture 4): $\overline{E} = E \cup E'$, where E' is the set of **limit points** of E (points x such that every neighborhood of x contains a point of E different from x).

So the closure takes E and "adds in" all the points that E accumulates toward.

Interior

Definition. The **interior** of a set $E \subseteq X$ is the largest open set contained in E:

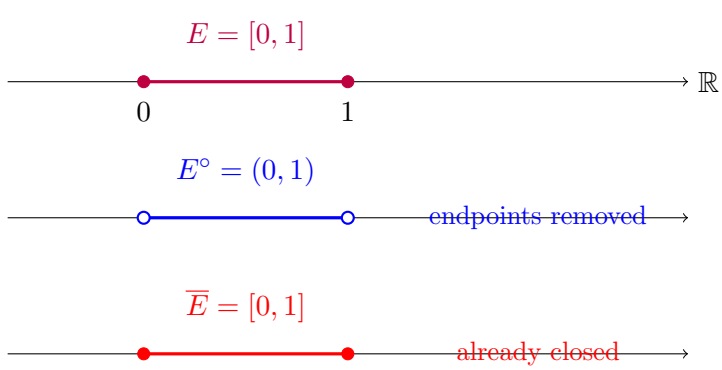
$$E^\circ = \bigcup \{U \subseteq X : U \text{ is open and } U \subseteq E\}.$$

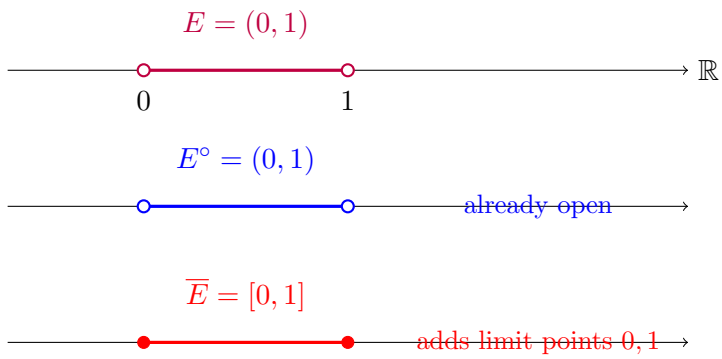
Equivalent characterization: $x \in E^\circ$ if and only if there exists an open neighborhood U of x with $U \subseteq E$. In a metric space:

$$x \in E^\circ \iff \text{there exists } \delta > 0 \text{ such that } B(x, \delta) \subseteq E.$$

So the interior strips away the "boundary" — the points that are in E but have no room around them.

Visual Comparison





Interior and closure are dual. Just as open and closed sets are complements, interior and closure are related by:

$$(X \setminus E)^\circ = X \setminus \overline{E}, \qquad \overline{X \setminus E} = X \setminus E^\circ.$$

The interior "shrinks" E to its open core; the closure "grows" E to include its boundary. Key relationships:

Property	**Interior** E°	**Closure** $\overline{E}$
Definition	Largest open set $\subseteq E$	Smallest closed set $\supseteq E$
Characterization	$\exists\, B(x, \delta) \subseteq E$	every nbhd of x meets E
Always satisfies	$E^\circ \subseteq E$	$E \subseteq \overline{E}$
E is open $\iff$	$E^\circ = E$	—
E is closed $\iff$	—	$\overline{E} = E$

1.4 Question 4: Connectedness

Question: Define connectedness.

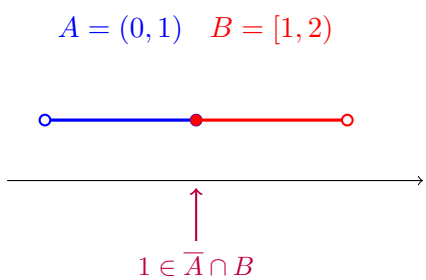
Separated Sets (Lecture 6)

Definition. Two subsets $A, B \subset X$ are **separated** if

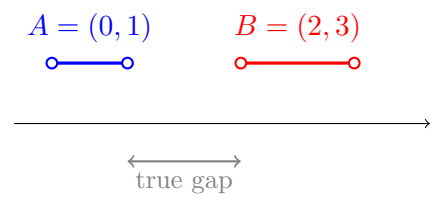
$$\overline{A} \cap B = \emptyset \qquad \text{and} \qquad A \cap \overline{B} = \emptyset.$$

This is stronger than just $A \cap B = \emptyset$ (disjoint). Separated means neither set contains nor "touches" limit points of the other. The sets must be truly "apart" — no point of one set is a limit point of the other.

Disjoint but NOT separated



Separated



Connected Sets (Lecture 6)

Definition. A set $E \subset X$ is **connected** if it is *not* the union of two nonempty separated sets. That is, there do *not* exist nonempty A, B with $A \cup B = E$, $\overline{A} \cap B = \emptyset$, and $A \cap \overline{B} = \emptyset$.

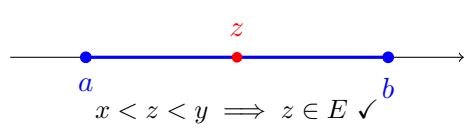
Intuitively: E is connected if you cannot split it into two pieces that are truly apart from each other. It is "all one piece."

Characterization in $\mathbb{R}$ (Lecture 6)

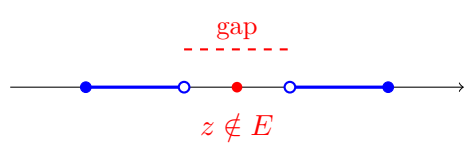
Theorem. $E \subset \mathbb{R}$ is connected if and only if: for all $x, y \in E$ with $x < z < y$, we have $z \in E$.

In other words, connected subsets of $\mathbb{R}$ are exactly the **intervals** (possibly unbounded, possibly open/closed/half-open). A connected subset of $\mathbb{R}$ has no "gaps."

Connected



Not connected



Proof approach (Lecture 6).
($\Rightarrow$) **Connected $\implies$ no gaps:** Contrapositive. If there is a gap ($x, y \in E$ but $z \notin E$ with $x < z < y$), split E into $A = E \cap (-\infty, z)$ and $B = E \cap (z, \infty)$. Since $z \notin E$, we get $A \cup B = E$. Both are nonempty ($x \in A$, $y \in B$). Since $\overline{A} \subseteq (-\infty, z]$ and $B \subseteq (z, \infty)$, the sets are separated. So E is disconnected.
($\Leftarrow$) **No gaps $\implies$ connected:** Contrapositive. If $E = A \cup B$ with A, B nonempty and separated, pick $x \in A$, $y \in B$ with $x < y$. Let $z = \sup(A \cap [x, y])$. This z (or a point near it) turns out to be in E but creates a contradiction with the separation condition, producing a gap.
The key technique in ($\Leftarrow$): use the **supremum** to find the "boundary" between A and B, then show this boundary point cannot belong to either set.

1.5 Question 5: Convergence

Question: Define convergence.

Convergence of a Sequence (Lecture 4)

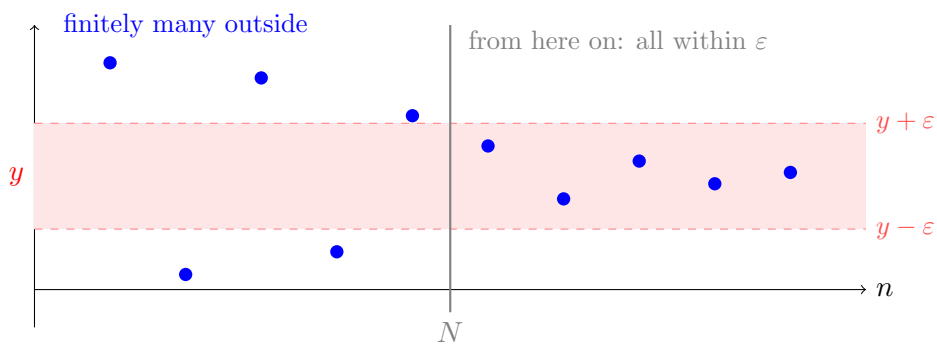
Definition. A sequence (x_n) in a topological space X **converges** to $y \in X$ if for every neighborhood U of y, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n \in U$.

In a metric space (X, d), this becomes the familiar ε-N definition:

$(x_n) \to y$ if and only if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies d(x_n, y) < \varepsilon.$$

Intuition: No matter how small a ball $B(y, \varepsilon)$ you draw around y, eventually (past some index N) *all* terms of the sequence land inside that ball. Finitely many terms may be outside, but from some point on, they all stay close.



Key Points

- **Limit points vs. limits of sequences (Lecture 4):** If p is a limit point of E, then every neighborhood of p contains *infinitely many* points of E (proved in Lecture 4). This means we can construct a sequence in E converging to p.
- **N depends on ε:** Smaller ε (tighter tolerance) generally requires a larger N (more terms before the sequence settles in). The definition requires that *for every* ε, *some* N exists.
- **Convergence in $\mathbb{R}^n$:** A sequence $(\vec{x}_n)$ in $\mathbb{R}^n$ converges to $\vec{y}$ if and only if each coordinate converges: $\vec{x}_n \to \vec{y}$ iff $x_n^{(i)} \to y^{(i)}$ for all $i = 1, \ldots, n$. (This follows from the equivalence of the product topology and the metric topology on $\mathbb{R}^n$, proved in Lecture 4.)

Common proof pattern. To show $(x_n) \to y$:

1. **Let $\varepsilon > 0$ be given.** (You are proving a "for all ε" statement.)
2. **Find** N (usually in terms of ε) such that $d(x_n, y) < \varepsilon$ for $n \geq N$. Often this involves solving an inequality like $\frac{1}{n} < \varepsilon$, giving $N > \frac{1}{\varepsilon}$.
3. **Verify:** For $n \geq N$, compute or bound $d(x_n, y)$ and show it is $< \varepsilon$.

The Archimedean property (Lecture 2) guarantees that N exists: for any $\varepsilon > 0$, there is $N \in \mathbb{N}$ with $N > 1/\varepsilon$.

1.6 Question 6: The Cauchy Criterion

Question: State the Cauchy criterion and explain what it means.

Statement (Lecture 9)

Definition. A sequence $\{s_n\}$ in a metric space (X, d) is a **Cauchy sequence** if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$,

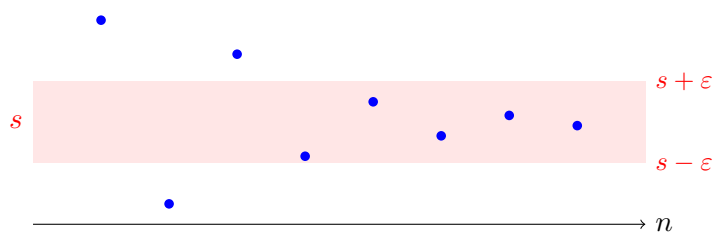
$$d(s_m, s_n) < \varepsilon.$$

Cauchy Criterion. In $\mathbb{R}$ (or $\mathbb{R}^n$), a sequence converges if and only if it is Cauchy.

What It Means

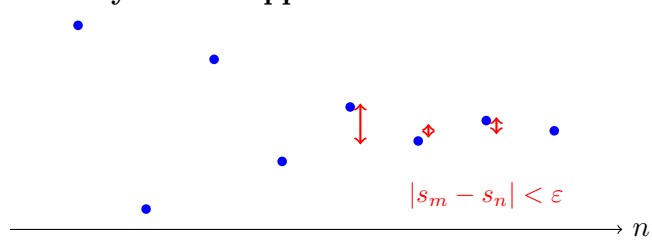
The standard definition of convergence ($s_n \to s$) requires you to *know the limit* s in advance. The Cauchy criterion removes this requirement: instead of checking that terms get close to some target, you check that *terms get close to each other*.

Convergence: terms approach s



Must know s!

Cauchy: terms approach each other



No limit needed!

Why This Matters

1. **Practical:** For many sequences (especially series), computing the exact limit is hard or impossible. The Cauchy criterion lets you prove convergence anyway.

2. **For series (Lecture 9):** If $s_n = \sum_{k=0}^{n} a_k$ are partial sums, the Cauchy condition becomes: for every $\varepsilon > 0$, there exists N such that for all $m > n \geq N$,

$$\left| \sum_{k=n+1}^{m} a_k \right| < \varepsilon.$$

This says: the "tail sums" become arbitrarily small.

3. **Completeness:** The fact that Cauchy $\iff$ convergent is a property of $\mathbb{R}$, not of all metric spaces. A metric space where every Cauchy sequence converges is called **complete**. The rationals $\mathbb{Q}$ are *not* complete: the sequence $1, 1.4, 1.41, 1.414, \ldots$ is Cauchy in $\mathbb{Q}$ but converges to $\sqrt{2} \notin \mathbb{Q}$. This connects back to Question 1: $\mathbb{R}$ is the *completion* of $\mathbb{Q}$.

Key corollary: If $\sum a_n$ converges, taking $m = n+1$ in the Cauchy condition gives $|a_{n+1}| < \varepsilon$ for large n, so $a_n \to 0$. The converse is false: $\sum 1/n$ diverges even though $1/n \to 0$ (the harmonic series).

1.7 Question 7: Sequential Compactness

Question: Define sequential compactness and its broader meaning.

Definition

A set K is **sequentially compact** if every sequence in K has a subsequence that converges to a point in K.

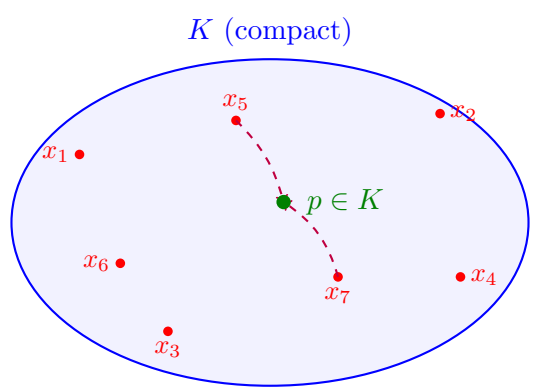
Relation to Compactness (Lecture 5)

In $\mathbb{R}^n$, the Heine–Borel theorem gives three equivalent characterizations:

(1)	K is **closed and bounded**
(2)	K is **compact**: every open cover has a finite subcover
(3)	Every **infinite subset** of K has a limit point in K

Condition (3) is essentially sequential compactness. If every infinite subset has a limit point in K, then given any sequence (x_n) in K:

- If the sequence takes infinitely many distinct values, that infinite subset has a limit point $p \in K$, and we can extract a subsequence converging to p.
- If the sequence takes only finitely many values, some value repeats infinitely often, giving a constant (hence convergent) subsequence.



No matter how the sequence bounces around,
a subsequence must converge to some $p \in K$.

Broader Meaning

Sequential compactness means a compact set is "inescapable": any infinite process (sequence) that stays in K must eventually cluster somewhere in K. You cannot have a sequence that keeps running away or spreading out.

This is why compactness is so powerful:

- **Extreme Value Theorem:** Continuous functions on compact sets attain max/min (the optimizing sequence must have a convergent subsequence).

- **Uniform continuity:** Continuous functions on compact sets are uniformly continuous (used in Lecture 6).
- **Bolzano–Weierstrass:** Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence (the Weierstrass theorem from Lecture 5).

1.8 Question 8: Convergence of a Sequence and a Series

Question: Define convergence of a sequence and of a series.

Convergence of a Sequence (Lecture 4)

Definition. A sequence (x_n) in a metric space (X, d) **converges** to $y \in X$, written $x_n \to y$, if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies d(x_n, y) < \varepsilon.$$

Convergence of a Series (Lecture 9)

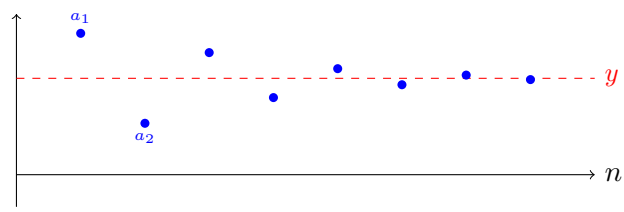
A series is *not* a new object — it is a sequence of partial sums.

Definition. Given a sequence $\{a_n\}$, the **partial sums** are

$$s_n = \sum_{k=0}^{n} a_k = a_0 + a_1 + \cdots + a_n.$$

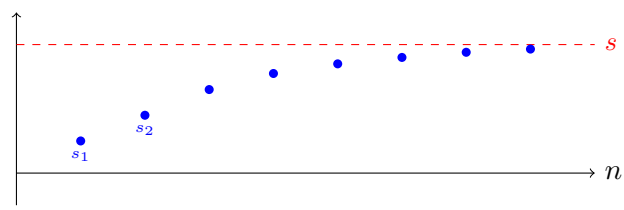
The series $\sum_{n=0}^{\infty} a_n$ **converges** to s if the sequence of partial sums converges: $s_n \to s$.

Sequence: $a_n \to y$



The *terms* a_n approach y

Series: $s_n = \sum_{k=0}^{n} a_k \to s$



The *cumulative sums* s_n approach s

Key distinction. "$a_n \to 0$" and "$\sum a_n$ converges" are *different* statements.

- If $\sum a_n$ converges, then $a_n \to 0$ (necessary condition, proved via the Cauchy criterion with $m = n + 1$).
- The converse is **false**: $a_n = 1/n \to 0$, but $\sum 1/n = \infty$ (harmonic series diverges).

So $a_n \to 0$ is necessary but not sufficient for convergence of the series.

1.9 Question 9: Convergent Sequences are Bounded

Question: Prove that convergent sequences are bounded.

Statement

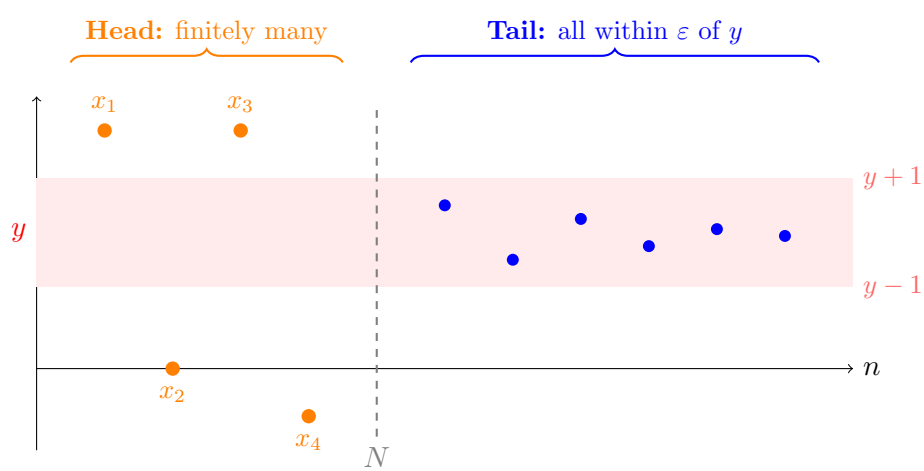
Theorem. If $(x_n) \to y$ in a metric space (X, d), then the sequence (x_n) is bounded: there exists $M > 0$ such that $d(x_n, y) \leq M$ for all n.

Concepts Involved

- **Convergence (Lecture 4):** $x_n \to y$ means for every $\varepsilon > 0$, there exists N such that $d(x_n, y) < \varepsilon$ for all $n \geq N$.
- **Boundedness (Lecture 4):** A set E is bounded if $E \subseteq B(x, M)$ for some $x \in X$ and $M > 0$.

Proof Approach

The idea is simple: convergence controls the *tail* (all terms past some N), and the *head* (finitely many initial terms) is automatically bounded.



1. **Use** $\varepsilon = 1$ in the convergence definition: there exists N such that $d(x_n, y) < 1$ for all $n \geq N$.
2. **The tail is bounded:** All terms with $n \geq N$ satisfy $d(x_n, y) < 1$.
3. **The head is finite:** The terms $x_1, x_2, \ldots, x_{N-1}$ are finitely many, so $\max\{d(x_1, y), \ldots, d(x_{N-1}, y)\}$ is finite.
4. **Combine:** Let $M = \max\{d(x_1, y), \ldots, d(x_{N-1}, y), 1\}$. Then $d(x_n, y) \leq M$ for all n.

Why this works. The key insight is the split into "finitely many" and "eventually all." Convergence gives you control over the tail. The head is automatically tamed because *any finite set is bounded*. Taking the max combines both.
This is a very common pattern in analysis: control the tail by the definition, and handle the (finite) head separately.

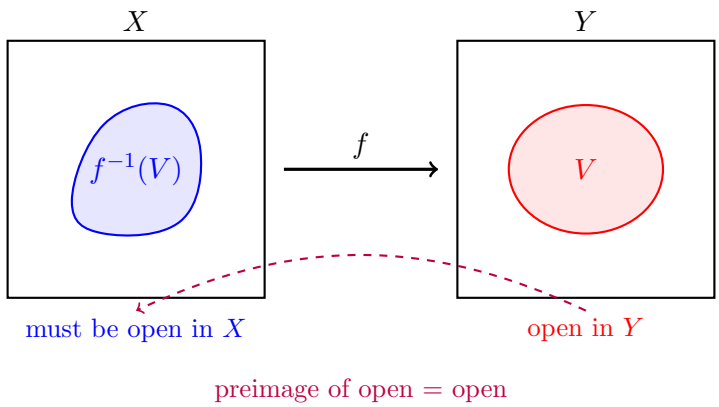
1.10 Question 10: Continuity

Question: Define continuity.

The Topological Definition (Lecture 6)

Definition. A function $f : X \to Y$ between topological spaces is **continuous** if for every open set $V \subseteq Y$, the preimage $f^{-1}(V)$ is open in X.

This is the most general definition. It says: open sets in the codomain "pull back" to open sets in the domain.

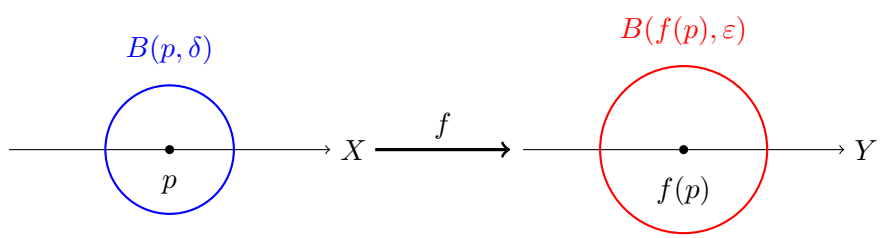


Three Equivalent Formulations (Lecture 6)

1. **Basis formulation:** Given a basis $\mathcal{B}$ for the topology on Y, f is continuous if and only if $f^{-1}(B)$ is open in X for every $B \in \mathcal{B}$. (You only need to check basis elements, not all open sets.)

2. **Local formulation:** f is **continuous at a point** p if for every open $V \ni f(p)$ in Y, there exists open $U \ni p$ in X with $f(U) \subseteq V$. Then f is continuous (globally) if and only if f is continuous at every point.

3. **ε-δ formulation (metric spaces):** f is continuous at p if for every $\varepsilon > 0$ there exists $\delta > 0$ such that
$$d_X(x, p) < \delta \implies d_Y(f(x), f(p)) < \varepsilon.$$



Points within δ of p map to points within ε of $f(p)$

Intuition. Continuity means "small changes in input produce small changes in output." The ε-δ definition makes this precise: you choose how close you want outputs to be (ε), and I find how close inputs need to be (δ).

The quantifier order matters:

$$\underbrace{\forall \varepsilon > 0}_{\text{you pick tolerance}} \qquad \underbrace{\exists \delta > 0}_{\text{I find a response}} \qquad \underbrace{\forall x : d(x,p) < \delta \implies d(f(x), f(p)) < \varepsilon}_{\text{guarantee}}$$

Crucially, δ is allowed to depend on both ε *and* p. This is what distinguishes continuity from **uniform continuity** (Question 11), where δ must work for all p simultaneously.

1.11 Question 11: Continuous Functions on Compact and Connected Sets

Question: What is the effect of a continuous function on a compact/connected set?

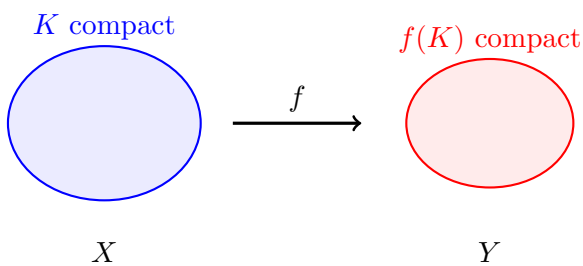
The Main Theorems (Lecture 6)

Theorem. If $f : X \to Y$ is continuous and $K \subseteq X$ is **compact**, then $f(K)$ is compact.

Theorem. If $f : X \to Y$ is continuous and $E \subseteq X$ is **connected**, then $f(E)$ is connected.

In short: *continuous functions preserve compactness and connectedness.*

Compactness Is Preserved



Proof idea (Lecture 6): Start with an open cover $\{V_\alpha\}$ of $f(K)$. Pull back: $\{f^{-1}(V_\alpha)\}$ is an open cover of K (by continuity). Extract a finite subcover (by compactness of K). Push forward: the corresponding V_{α_i}'s cover $f(K)$.

Consequences of Preserving Compactness

1. Extreme Value Theorem (Lecture 6). If $f : K \to \mathbb{R}$ is continuous and K is compact, then f attains its maximum and minimum.

Why: $f(K)$ is compact in $\mathbb{R}$, hence closed and bounded (Heine–Borel). Bounded means $\sup f(K)$ and $\inf f(K)$ are finite. Closed means $f(K)$ contains its limit points, so $\sup f(K) \in f(K)$ and $\inf f(K) \in f(K)$.

2. Uniform Continuity (Lecture 6). If $f : K \to Y$ is continuous and K is compact, then f is uniformly continuous.

Recall: f is uniformly continuous if δ can be chosen independent of the point p:

$$\forall \varepsilon > 0 \;\; \exists \delta > 0 \;\text{ such that }\; d(p,q) < \delta \implies d(f(p), f(q)) < \varepsilon \quad \forall p, q \in K.$$

Why compactness helps: At each point p, continuity gives a δ_p. Cover K by balls $B(p, \delta_p/2)$, extract a finite subcover, and take $\delta = \frac{1}{2}\min(\delta_{p_1}, \ldots, \delta_{p_n})$.

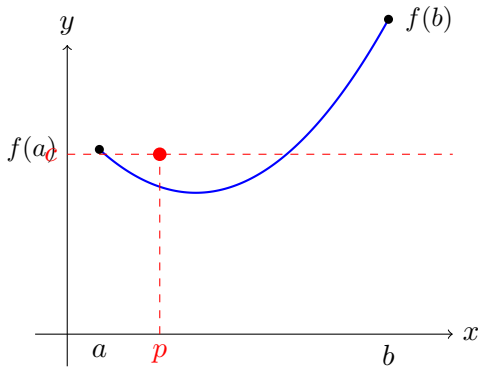
Connectedness Is Preserved

Proof idea (Lecture 6): If $f(E)$ were disconnected as $A \cup B$ (separated), then $f^{-1}(A) \cap E$ and $f^{-1}(B) \cap E$ would separate E, contradicting the connectedness of E.

Consequence: Intermediate Value Theorem

IVT (Lecture 6). If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) < c < f(b)$, then there exists $p \in (a, b)$ with $f(p) = c$.

Why: $[a, b]$ is connected. So $f([a, b])$ is connected in $\mathbb{R}$. Connected subsets of $\mathbb{R}$ are intervals (have no gaps). Since $f(a)$ and $f(b)$ are in $f([a, b])$, every value between them is too.



f continuous on $[a, b]$ (connected)
$\implies$ $f([a, b])$ connected (no gaps) $\implies$ c is hit

Summary table.

K **is...**	$f(K)$ **is...**	**Consequence**
Compact	Compact	Extreme Value Thm, Uniform Continuity
Connected	Connected	Intermediate Value Theorem

What compactness does NOT preserve (Lecture 6): On a non-compact set E, a continuous function can be (1) unbounded, (2) bounded but miss its sup, or (3) not uniformly continuous. All three pathologies are eliminated by compactness.